

# Muhammad Ahmad Amin

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## Summary

Computer Science student at NUST with experience building full-stack MERN applications, data-driven analytics tools, and contributing to open-source projects. Solved 410+ algorithmic problems and interested in software engineering, data science, and scalable web applications.

## Education

**BS Computer Science** | National University of Sciences and Technology - NUST (2024 - 28)

## Skills

**Programming Languages** | Java, Python, JavaScript

**Data Science** | Python, NumPy, Pandas, Matplotlib, Seaborn

**Frontend Development** | HTML, CSS, JavaScript, React.js, TailwindCSS

**Backend Development** | Node.js, Express.js, REST APIs, Postman (tool)

**Databases** | MySQL (Relational Database/SQL), MongoDB (noSQL), Mongoose

**Version Control System** | Git, GitHub

## Projects

- **FactArena** | Full-Stack AI-Judged 1v1 Debate Platform (Argument Settler) [Live Demo](#) | [GitHub](#)
  - Built a **competitive AI debate** platform where two players argue a topic in turns and an **impartial AI judge** analyzes the full transcript to declare a winner with individual scores and reasoning.
  - Engineered a **turn-based** argument system with a synchronized countdown timer that automatically submits the debate transcript to an **LLM judge** on expiry.
  - Built with React, Node.js, Express, and integrated Groq API for impartial judging. **Deployed** on Netlify and Render.
- **Linracy** | Full Stack MERN Social Media Platform [Live Demo](#) | [GitHub](#)
  - Built a **full-stack** social media platform with user **JWT-based authentication**, profiles, photo uploads via Cloudinary and **post management** using React.js, Node.js, Express and MongoDB (**MERN Stack**).
  - Built a sleek, **responsive** user interface optimized for both desktop and mobile using modern frontend design practices.
  - **Deployed** frontend on Netlify and backend on Render.
- **O3Scope** | Ozone Analysis, Visualization and Prediction Using Nasa Dataset [Hex App](#) | [GitHub](#)
  - Built an **interactive Hex app** visualizing global and city-level ozone trends (2012–2026) with risk zones.
  - Developed linear regression **predictions for 2027 ozone levels** using Hex AI-assisted analysis.
  - Processed **NASA OMPS HDF5 datasets** and created **dynamic visualizations** with Matplotlib & Cartopy.

## Open Source Contributions

**hnn-core** | Human Neocortical Neurosolver

[GitHub](#)

**Merged PR:** Improved GUI usability by refactoring button implementation to standard Widget for consistency.

## Problem Solving & Competitive Programming

[Leetcode Profile](#)

Solved 410+ Data Structures and Algorithms Problems on Leetcode and ranked in the top 3% globally among 15M+ users.

## Hackathons

**Hex-a-thon** | Built O3Scope, an interactive ozone data analysis, visualization and prediction tool using NASA datasets